

F6: Three convergent evidence streams support the same boundary conclusion
(Each stream is independently consistent, not an independent causal proof)

Study 1: Meal Counterfactual Audit -- Evidence Boundary

Orthogonalization
Leaky partial corr: 0.519
Causal partial corr: 0.027
Deployable features lack
meal-specific signal

Glucose-defined arm
 $\text{delta_theta} = -35.2 \text{ mg/dL}$
(inverted vs $+22.4$ observed)
Label is mean-reversion marker,
not causal meal driver

Wearable pre-rise arm
 $\text{delta_theta} = -0.21$ (dose 1.0)
 $= -0.79$ (dose 6.25)
Scales more negative at
higher dose; sign persists

Ruling: not leakage headroom
C-C' gap is fully unreachable

Ruling: not encoding artefact
Lead preservation confirmed (Step 2)

Ruling: not dose artefact
Dose-matched sweep stays negative

BOUNDARY CONCLUSION
AI-READI cannot support a meal counterfactual in any constructible label.
Observed glucose rise: $\text{delta_obs} = +22.4 \text{ mg/dL}$ at 60 min.
Both constructible labels fail sign agreement (not under-dose, not dead channel, not encoding).
Next required step: exogenous meal annotation (e.g., T1DEXI or CGMacros dataset).